

Mika Be

APPLIED AI | SOFTWARE ENGINEERING

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EDUCATION

Rutgers University - New Brunswick | B.S. Electrical and Computer Engineering, CS minor

May 2028

GPA 3.838/4.000 | Dean's List, all four completed semesters

Honors IEEE-HKN member | BCA Computer Science Departmental Award, 1 of 3 recipients

Coursework Data Structures; Digital Logic Design; Computer Architecture & Assembly and Linear Systems & Signals (in progress)

RESEARCH

Kuiper TTS | Undergraduate Researcher and Co-author

2025-Present

Python | *PyTorch Lightning* | *Piper/VITS* | *ONNX* | *LaTeX*

- Built a reproducible GPU training and ONNX export pipeline for Piper/VITS voices; resolved dependency incompatibilities and validated cross-platform inference.
- Implemented LaTeX-to-speech rules for fractions, derivatives, grouping, and operator pacing; linked source text, settings, audio, and duration in traceable manifests.
- Coordinated manuscript and reproducibility work for an 18-member team; co-authored a paper awarded 2nd Place Best Full Paper at IEEE ISEC 2026.

SELECTED SOFTWARE PROJECTS

Room Notes | Solo Developer, in development

2026-Present

Flutter | *Dart* | *Firebase* | *Drift/SQLite*

- Built a cross-device notebook where typed text, ink, equations, code, images, and diagrams share a stable 900 x 1200 page model.
- Designed local-first persistence and conflict-aware synchronization for notebooks and media, with automated regression coverage for edits and recovery.

Knowsy Breadboard Lab | Technical Lead, three-person team

2025-Present

JavaScript | *circuit simulation* | *automated testing*

- Led an interactive breadboard lab with wiring validation, digital simulation, and inspectable netlist output.
- Implemented modified nodal analysis over complex admittances and frequency sweeps for voltage, phase, and Bode-plot calculations.

Room of Days | Solo Developer, active development

2025-Present

Flutter | *Dart* | *local-first architecture* | *Firebase*

- Built a planning and reflection app for Android, iOS, and web with an offline core, validated persistence, backups, and optional cloud sync.

EXPERIENCE

Schieber Research Group | Software Developer

2023-2024

- Built and documented a React/Vite research prototype backed by Appwrite, translating study requirements into reusable interfaces.

Starbucks | Barista

2023-2025

TECHNICAL SKILLS

Languages Python, JavaScript, Java, Dart, C/C++, SQL, HTML/CSS

AI and research PyTorch Lightning, Piper/VITS, ONNX, GPU training, LaTeX parsing, reproducible experiments

Software Flutter, Firebase, Drift/SQLite, React, Git, automated testing, local-first sync